A Visual-World Study of Processing Hungarian Pre-Verbal Focus Constructions

Gábor Parti and Yu-Yin Hsu*
gabor.parti@connect.polyu.hk, yyhsu@polyu.edu.hk*
The Hong Kong Polytechnic University

How languages express contrast and exhaustivity has major implications for how we understand the syntax-semantics interface. This study revisits the debate on the interpretation of Hungarian focus constructions using eye tracking with the visual-world paradigm. Specifically, we examine two theoretical inquiries: how strongly Hungarian native speakers interpret phrases in the pre-verbal focus domain *exhaustively*, and whether *contrastive but non-exhaustive* readings exist, as well as from a cognitive perspective, whether different focus types require different processing efforts during language comprehension.

Constituents in a sentence can be focused for various communicative purposes, such as emphasis, question answering, and contradiction. For example, if one wants to emphasize the object of (1a), the typical focus construction would be like (1b), called the **pre-verbal focus (PVF)**, in which the focused element is moved to an immediately pre-verbal position, and it bears eradicating stress (Kornai & Kálmán, 1988). If the verb has a verbal prefix (VPX, e.g., *meg* in (1)), they will detach and get left behind. An intuitive reading of (1b) could be "*Anna ate the BANANA [and nothing else]"*; in this sense, PVF sentences seem to be equal to (1c) with an exhaustive focus operator *csak* 'only'. However, we also notice that if the utterance (1b) is amended with certain phrases, e.g. *többek között* 'amongst others' (1d), it is not only perfectly grammatical but also expresses a contrastive sense. In Hungarian linguistic work, <u>semantic exhaustivity as a core feature of PVF has been assumed and debated for over 40 years</u>. A fundamental question is: *Does the exhaustive denotation of PVF arises from the syntax-semantic interface, or from pragmatic/contextual factors*? Within the generative framework, several studies argue that exhaustivity is inherent of PVF (Szabolcsi, 1981; É. Kiss, 1998; Kenesei, 2006), while others advocate that exhaustive interpretation emerged from discourse-pragmatics, or conversational implicature (Wedgwood et al., 2006; Balogh, 2013, Káldi et al., 2016).

To address this theoretical debate, and in light with recent experimental studies (Onea & Beaver, 2009; Gerőcs et al., 2014; Káldi, 2021), we designed a visual-world eye-tracking experiment (n = 55; MeanAge = 35, female = 32) to differentiate interpretations associated with the PVF structure. Specifically, we were curious about Hungarian speakers' dominant interpretation of the unmodified, simple PVF sentence (1b), in comparison with the *only*-exhaustive (1c), and the contrastive conditions (1d). We used a Sentence–Picture Verification (SPV) task (Fig. 1), in which participants read a sentence in a self-paced reading (SPR) manner and were asked to indicate one correct meaning-representation between two visual stimuli. Materials constituted of 54 sentences (18 items × 3 conditions), and 54 pairs of images with relevant single and multiple objects (Fig. 2), and 54 filler trials. Using PCIbex (Schwarz & Zehr, 2021), we collected reading times (RT) during SPR, and eye-tracking data (dwell time) during the SPV choice, recording the time spent looking at the target images. It is assumed that less straightforward/uncommon choices would elicit longer reading time and more frequent looks, reflecting higher cognitive load (Gidlöf et al. 2013).

The behavioural results showed that participants overwhelmingly interpreted the unmodified PVF sentences like exhaustive *only*-sentences, choosing single-object images (A), but in the contrastive condition (*among other*-sentences) they dominantly preferred the multi-object images (B) (Table 1).

Conditions significantly influence sentences' overall RTs: exhaustive sentences were read faster than unmodified (p<.0001) and contrastive sentences (p<.05). Pairwise comparisons showed that *only*-sentences were read especially fast when compared to unmodified sentences when people chose A (p=.0004). During word-by-word processing, in region corresponding to 'only', participants read this region significantly quicker than in unmodified condition (p=.0003); this same pattern was found in the object region (p=.0184), and the sentence-final region (p<.0001). Additionally, while other pairwise comparisons were insignificant, the sentence-final region was read faster when readers selected A in the exhaustive condition, i.e., explicit exhaustive readings were facilitated. The task also influenced the reaction times of choice. Effects were found on the Choice (p=.001), the Condition (p=.018) and in their interaction (p=.029): Decision-making was quicker in the exhaustive condition than in the contrastive condition (p=.005) regardless of selecting A or B (p*s* >.06), but it was not significantly different from the unmodified condition (p=.144). <u>In short, RT results show that explicit exhaustive focus denotation facilitated the reading</u>

comprehension, unlike the unmodified condition; however, to make a decision, the RTs needed between exhaustive and unmodified was not significantly different.

We fitted a linear mixed-effects model to the total dwell time on images (the sum of all gaze episodes' durations), with Condition, Choice, and their interaction as main effects, and Participant and Item as random effects. The results (Table 2) revealed significant effects of all three factors. Interestingly, post-hoc Tukey tests showed that after reading exhaustive sentences, participants took significantly less time to choose A compared to B ($\beta = -0.4443$, $p = .0081$), whereas they spent significantly more time to select A after reading a contrastive sentence than an exhaustive sentence ($\beta = 0.1876$, $p = .0057$). This suggests that participants were faster to pick the expected choices in exhaustive and contrastive conditions, respectively. Crucially, no significant difference was observed in dwell time between selecting A or B for the unmodified PVF sentences ($\beta = -.2390$, $p = .539$), implying that participants might consider both representations equally plausible when making their choices.

Overall, our results suggest that Hungarian PVF constructions tend to be interpreted exhaustively, when no additional context is provided, although non-exhaustive interpretations remain possible. Our study reconciles long-standing theoretical debates by showing that PVFs show an exhaustivity bias, while remaining semantically underspecified during comprehension. The exhaustive bias seems to persist, but much weakened, when 'amongst others' is present, despite its licensing a contrastive reading in truth value.

(1) a *Anna* *meg-ette* *a banánt.*
Anna[NOM] VPX-eat.PST.3SG the banana-ACC
'Anna ate the banana.'

b *Anna* ***a BANÁNT*** *ette meg.*
Anna[NOM] the banana-ACC eat-PST.3SG VPX
'Anna ate the BANANA.'

c *Anna* ***csak*** ***a BANÁNT*** *ette meg.*
Anna[NOM] only the banana-ACC eat- PST.3SG VPX
'Anna ate only the BANANA.'

d *Anna* ***többek között*** ***a BANÁNT*** *ette meg.*
Anna[NOM] among others the banana-ACC eat- PST.3SG VPX
'Anna ate the BANANA, among other things.'

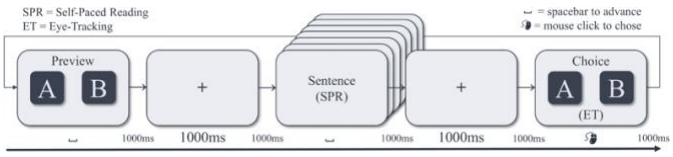

Figure 1. The design and procedure of the Sentence–Picture Verification task.

A B

Figure 2. Single-(A) and multi-object (B) target.

Condition	A % (N)	B % (N)
Exhaustive	95.1% (411)	4.86% (21)
Unmodified	96.1% (415)	3.94% (17)
Contrastive	37.7% (163)	62.3% (269)

Table 1. Choice type per condition.

Effect	Chisq	Df	Pr(>Chisq)
(Intercept)	10997.369	1	< 2.2e-16
condition	12.531	2	0.0019008
choice	11.845	1	0.0005781
condition × choice	12.292	2	0.0021419

Table 2. Chi-square tests of total dwell time during choice (ms).

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